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BACHELOR THESIS

Thesis Template in Typst

A Demonstration of the FHNW Thesis Template

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Abstract

This document demonstrates the FHNW thesis template for Typst. It walks through all available features – title page, table of contents, chapters, figures, tables, mathematical formulas, citations, custom environments, and draft-mode TODO markers – so you can start writing your own thesis.

Acknowledgments

I would like to thank my supervisor for their guidance throughout this project, as well as the Institute of Mobile and Distributed Systems at FHNW for providing the resources that made this work possible.

Table of Contents

1

Introduction

This thesis serves as a tutorial and demonstration of the FHNW thesis template for Typst. The template aims to simplify the thesis writing process by leveraging Typst's modern, fast typesetting capabilities.

Throughout this document you will find examples of citations [1], cross-references, mathematical formulas, figures, tables, and other essential elements commonly used in academic writing.

1.1. Motivation

Writing a thesis involves more than just content – presenting it in a clean, consistent, and professional layout is equally important. A good template lets you focus on the writing while guaranteeing a uniform appearance across the whole document.

1.2. Objectives

The objectives of this template are:

- Provide a clean, FHNW-branded layout out of the box.
- Keep configuration simple through a single `#show: thesis.with(...)` call.
- Remain flexible enough for Bachelor, Master, and project theses.

By the end of this tutorial you should be familiar with all of the template's features and ready to write your own thesis using Typst.

2

Background

This chapter introduces the foundational concepts that the remainder of the thesis builds upon. It is intentionally short – replace it with the relevant theory for your own work.

2.1. Typesetting with Typst

Typst is a modern, markup-based typesetting system designed as a fast and approachable alternative to LaTeX. It compiles documents in milliseconds, provides a clean scripting language, and ships with a growing package ecosystem.

2.2. Document Structure

A thesis typically consists of front matter (title page, abstract, acknowledgments, table of contents), the main chapters, and back matter (bibliography and appendices). This template handles all of these automatically once you pass your content to the `thesis` function.

See Section 1 for the introduction and Section 3 for the methodology that follows.

3

Methodology

This chapter outlines the approach taken in the thesis. Describe the methods, tools, and processes you used to address the research questions.

3.1. Research Design

Explain whether your work is experimental, analytical, or design-oriented, and justify the choice. Reference related work where appropriate, for example [2].

3.2. Tools and Technologies

List the primary tools used in the project:

- **Typst** – document preparation.
- **Git** – version control.
- **Python** – prototyping and data analysis.

3.3. Process

The following pipeline summarises the overall process. Each stage is detailed in the subsequent chapters.

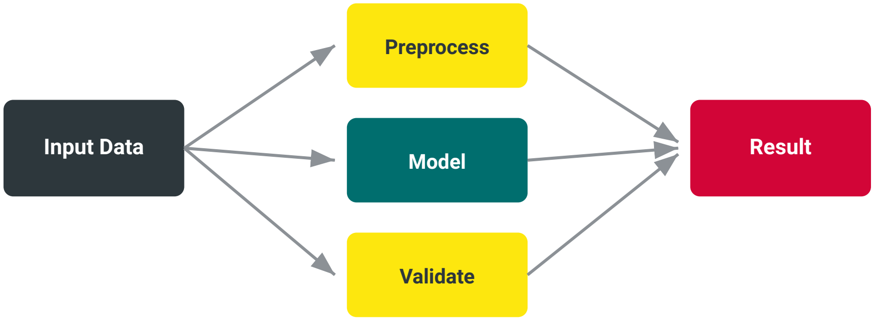

Figure 1: Overview of the project pipeline.

As illustrated in Figure 1, the workflow moves from input data through preprocessing and modelling to the final result.

4

Implementation

This chapter demonstrates every typographic feature the template offers. Use it as a reference while writing your own thesis.

4.1. Text Formatting

You can use **bold text** for emphasis, *italic text* for titles or foreign words, and `inline code` for technical terms. The template also supports SMALL CAPITALS for certain stylistic choices.

Footnotes are created with the `#footnote` function[1]. You can add them anywhere in your text.

For URLs you have several options:

- Direct URL: https://www.fhnw.ch
- Linked text: FHNW website

4.2. Lists and Enumerations

Bullet points are created naturally:

- First item
- Second item with sub-items:
  - Sub-item A
  - Sub-item B
- Third item

Numbered lists work similarly:

1. First step

[1] This is an example footnote. Footnotes are automatically numbered and appear at the bottom of the page.

2. Second step
3. Final step

4.3. Code Listings

Code blocks are formatted with syntax highlighting:

```
def fibonacci(n):
    """Calculate the nth Fibonacci number."""
    if n <= 1:
        return n
    return fibonacci(n - 1) + fibonacci(n - 2)
```

For inline code, simply use backticks: `let x = 42`.

4.4. Mathematical Formulas

Typst excels at mathematical typesetting. Here is Euler's identity:

$$e^{i\pi} + 1 = 0 \quad (1.)$$

More complex equations are displayed prominently:

$$\int_0^\infty e^{-x^2} \, \mathrm{d}x = \frac{\sqrt{\pi}}{2} \quad (2.)$$

You can also create aligned equations:

$$\begin{aligned} \sum_{k=1}^{n} k &= 1 + 2 + 3 + \ldots + n \\ &= \frac{n(n+1)}{2} \end{aligned} \quad (3.)$$

As shown in Equation 3, aligned equations are referenced like any other element.

4.5. Figures and Tables

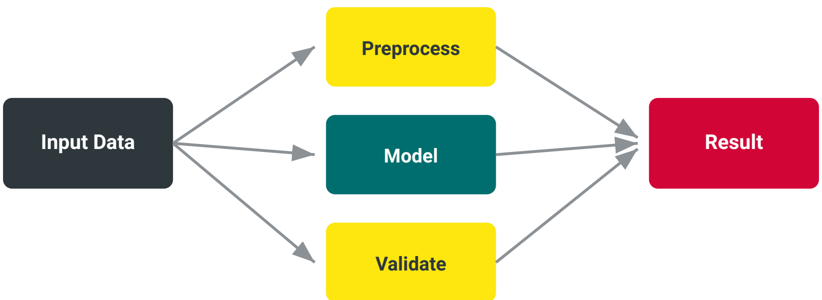

Figure 2: Example figure using the template's figure environment.

As shown in Figure 2, figures are automatically numbered and can be referenced throughout the document.

Tables are equally straightforward:

Language	**Type System**	**Performance**
Typst	Static	Fast
LaTeX	Macro-based	Slow
Markdown	None	Very Fast

Table 1: Comparison of different markup languages.

Table 1 demonstrates the three-column table format with different alignments.

4.6. Custom Environments

The template provides several custom environments for academic writing:

Definition 1. A **markup language** is a text-encoding system consisting of a set of symbols inserted in a text document to control its structure, formatting, or the relationship between its parts.

Theorem 1. For any typesetting system T, if T is both powerful and user-friendly, then T will eventually replace less user-friendly alternatives.

4.7. Algorithms

```
function binarySearch(arr, target):
    left = 0
    right = length(arr) - 1

    while left <= right:
        mid = (left + right) / 2
        if arr[mid] == target:
            return mid
        else if arr[mid] < target:
            left = mid + 1
        else:
            right = mid - 1

    return -1
```

Algorithm 1: Binary search algorithm.

4.8. Citations and References

Citations are handled naturally using the `@` symbol: [1] for a single citation or [2], [3] for multiple. The bibliography style is configured in the main document.

4.9. Draft Features

The template includes helpful TODO markers for draft versions:

TODO: Add more examples here.

TODO: Include performance benchmarks.

TODO: Verify this equation.

TODO: Add a proper reference.

TODO: Check grammar in this section.

TODO: Is this the best approach?

TODO: Remember to update this section.

4.10. Highlighting

Use the important function to highlight crucial information. In colored mode this appears as highlighted text; in non-colored mode it becomes italicized.

5

Evaluation

This chapter presents the evaluation of the approach described in Section 3. Discuss the experimental setup, the metrics used, and the results obtained.

5.1. Experimental Setup

Describe the datasets, hardware, and parameters used in the evaluation. Make the setup reproducible by another researcher.

5.2. Results

Present your results clearly, using figures and tables where appropriate. Compare your results against relevant baselines [3].

6

Discussion

In this chapter the results from Section 5 are interpreted and put into context. Discuss what the findings mean, why they turned out the way they did, and how they relate to the original objectives stated in Section 1.

6.1. Limitations

No study is without limitations. Be transparent about the boundaries of your work – this strengthens rather than weakens your thesis.

7 Conclusion

This thesis demonstrated the FHNW thesis template for Typst and walked through all of its features. Summarise the main contributions of your work and restate how they address the research questions posed in Section 1.

The template provides a clean, FHNW-branded layout, simple configuration, and the flexibility needed for Bachelor, Master, and project theses alike.

8

Future Work

Several directions remain for future work:

- Extend the evaluation to additional datasets and domains.
- Investigate alternative approaches not covered in this thesis.
- Improve performance through the optimisations discussed in Section 6.

These ideas build directly on the foundation laid by the present work and offer promising avenues for continued research.

Bibliography

[1] A. M. Turing, “Computing machinery and intelligence,” *Mind*, vol. 59, no. 236, pp. 433–460, 1950.

[2] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016.

[3] A. Vaswani and others, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, 2017.

A

Supplementary Material

This appendix contains additional material that supports the main text but would interrupt the flow if placed in the body chapters.

A.1 Data Samples

Include extended tables, raw data excerpts, or configuration files here. Appendix chapters are automatically lettered (A, B, C, …).

A.2 Additional Figures

Place figures that are supplementary rather than essential here.